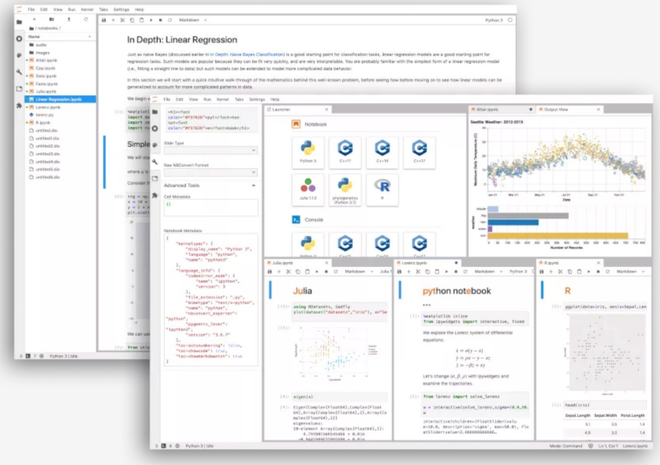


Set Up and Start Using Your Python Playground

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Graduate Institute of Data Science, Taipei Medical University, Taiwan.
Email: changyc@tmu.edu.tw

JupyterLab & Colab



JupyterLab: A Next-Generation Notebook Interface

JupyterLab is the latest web-based interactive development environment for notebooks, code, and data. Its flexible interface allows users to configure and arrange workflows in data science, scientific computing, computational journalism, and machine learning. A modular design invites extensions to expand and enrich functionality.

Try it in your browser | Install JupyterLab

Installing Jupyter

Get up and running on your computer

Project Jupyter's tools are available for installation via the Python Package Index, the leading repository of software created for the Python programming language.

This page uses instructions with pip, the recommended installation tool for Python. If you require *environment management* as opposed to just installation, look into conda, mamba, pipenv, and Homebrew.

JupyterLab

Install JupyterLab with `pip`:

```
pip install jupyterlab
```

Note: If you install JupyterLab with conda or mamba, we recommend using the conda-forge channel.

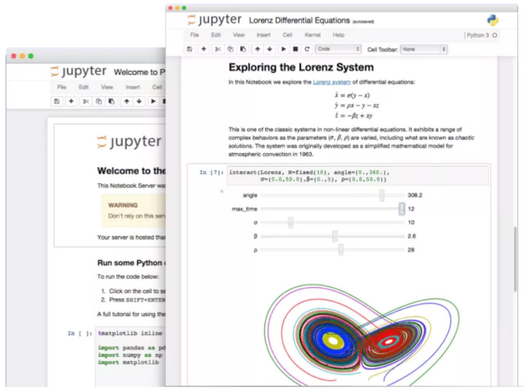
Once installed, launch JupyterLab with:

```
jupyter lab
```

Jupyter Notebook: The Classic Notebook Interface

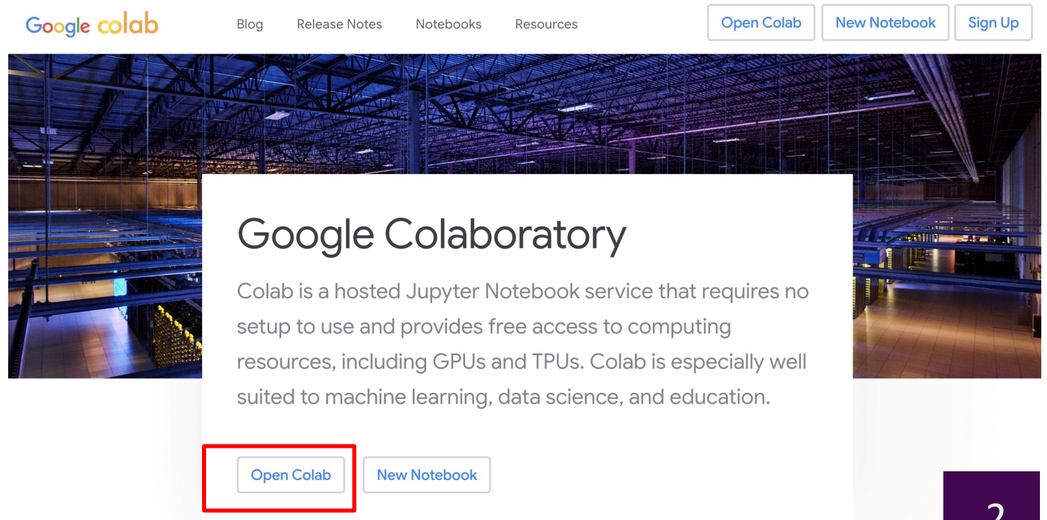
The Jupyter Notebook is the original web application for creating and sharing computational documents. It offers a simple, streamlined, document-centric experience.

Try it in your browser | Install the Notebook

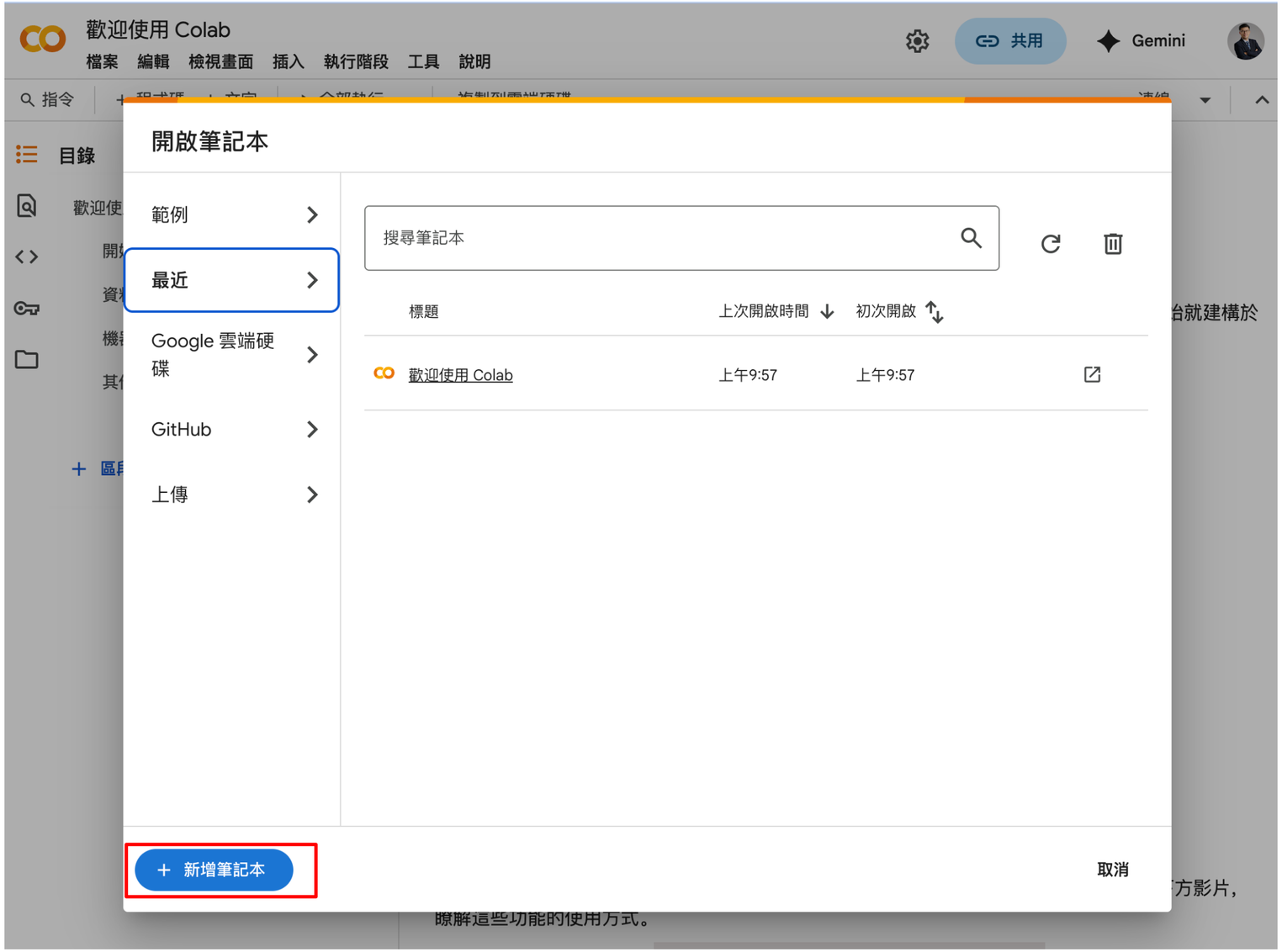


https://jupyter.org/





Colab (cont.)


歡迎使用 Colab
檔案 編輯 檢視畫面 插入 執行階段 工具 說明
共用
Gemini
指令
目錄
開啟筆記本
範例
最近
Google 雲端硬碟
GitHub
上傳
搜尋筆記本
標題
上次開啟時間
初次開啟
歡迎使用 Colab
上午9:57
上午9:57
新增筆記本
取消

Colab (cont.)

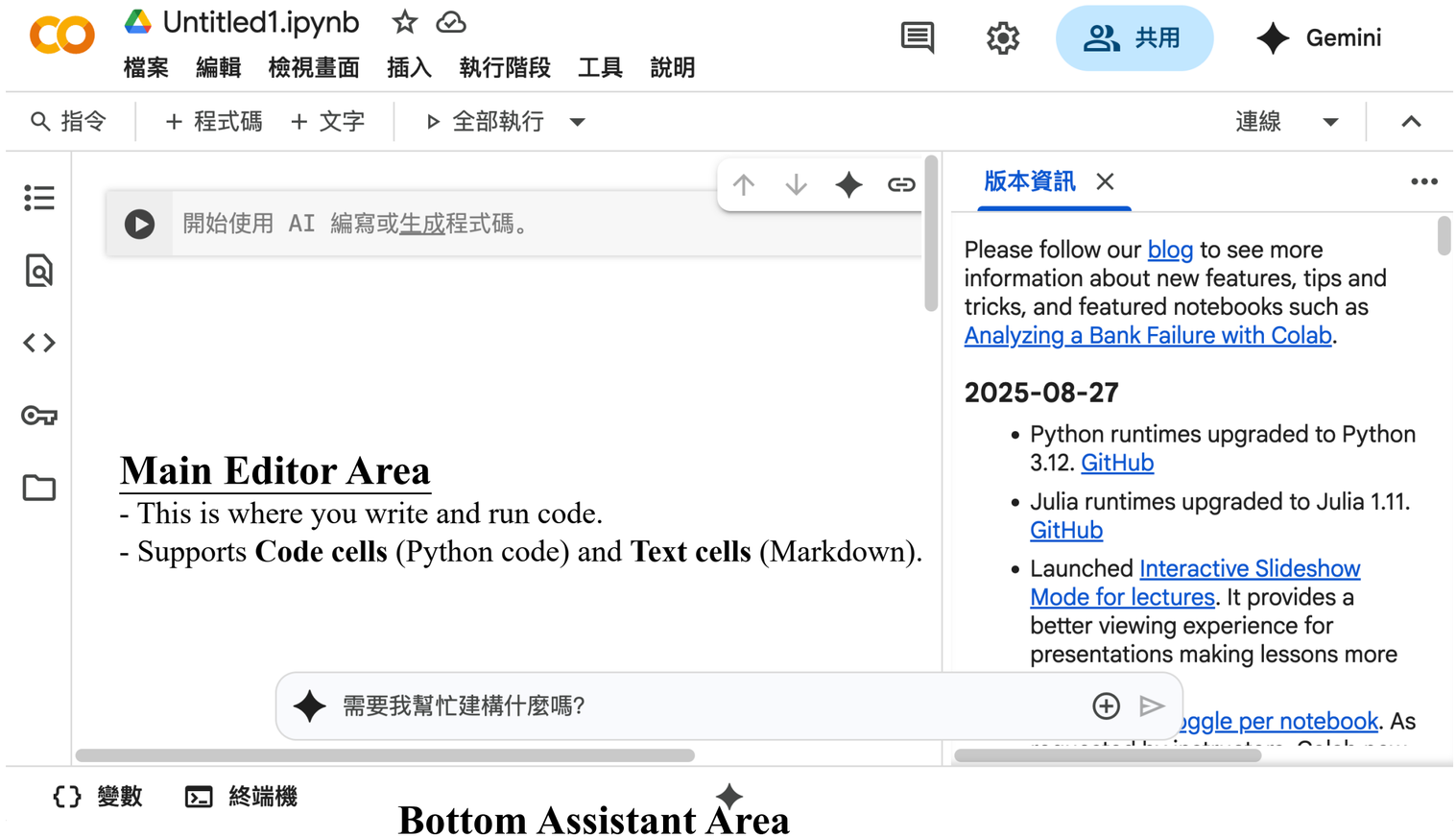
Menu Bar

File / Edit / View / Insert / Runtime / Tools / Help

Similar to Google Docs. You can create a new notebook, download code, adjust text formatting, insert code or text cells, and manage the runtime environment.

Toolbar

▶ **Run All**: Executes the entire notebook.
⏹ **Stop**: Interrupts the running code.
🔄 **Restart**: Restarts the Python runtime.
+ Code / + Text: Insert a new code cell or a text cell (for explanations, notes, etc.).



Left Sidebar

- **Folder Icon**: Browse or upload files (you can also mount Google Drive here).
- **Search Icon**: Search within the notebook.
- **Code Snippets Icon**: Insert sample snippets provided by Colab.
- **Extensions/Emoji Icon**: Extra add-ons or plugins.

Main Editor Area

- This is where you write and run code.
- Supports **Code cells** (Python code) and **Text cells** (Markdown).

Right Panel (Release Notes / Updates)

-Shows Colab's latest updates such as Python upgrades or package changes.
-Usually not needed for students, but instructors may check package versions here.

Bottom Assistant Area

-**"Need help with what to code?"**: AI code suggestions.

The MT Samples Dataset

MT Samples Dataset: Clinical Scenario Overview

From clinical encounter and physician documentation to structured medical transcription data

1 Clinical Encounter



- Chief Complaint
- History of Present Illness
- Physical Examination
- Initial Assessment

2 Physician Documentation



- Conversation / Dictation
- Examination / Test Results
- Diagnosis / Impression
- Plan / Dictation

Dictation / Note → Text Transcription

3 MT Samples Dataset

DE-IDENTIFIED TRANSCRIPTION

Chief complaint: chest pain

History of present illness: Patient is a 58-year-old male with intermittent chest pain for 2 days.

Assessment: possible angina

Plan: ECG and follow-up

Disposition: outpatient follow-up

- **description** — A brief summary of this record or patient encounter
- **medical_specialty** — The medical specialty label, e.g., Cardiology / Surgery
- **sample_name** — Sample title or record identifier
- **transcription** — Full clinical transcription text (complete document)
- **keywords** — Key medical terms or concepts in this record

A standardized, machine-readable medical transcription dataset

4 Fields in the Dataset

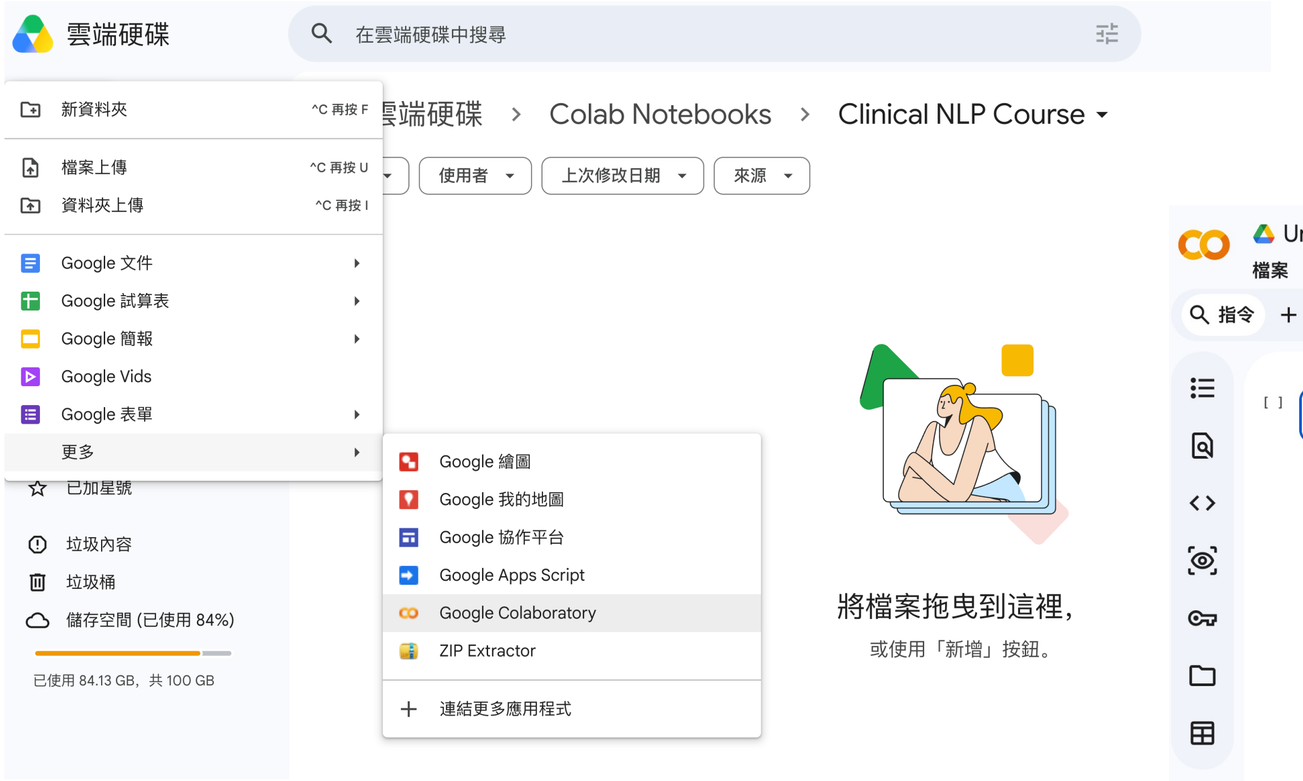
description	medical_ specialty	sample_ name	transcription	keywords
Chest pain, follow-up	Cardiology	Consult-10	Patient reports intermittent chest pain...	chest pain, ECG, angina

Each row represents one medical transcription case

What can we do with this dataset?

By organizing real clinical dictations and records into structured text data, we can analyze medical content, extract clinical insights, and build NLP models for clinical applications such as classification, summarization, and information extraction.

Create your iPython Notebook





The workflow for implementing ML model

- **Data Preparation**



  - Load Data using Dataframe
  - Data observation
  - Identify X (data representation) and y (label)
  - Data preprocessing

- **Model Training**
  - Split training data and testing data
  - Data Representation
  - Select the ML model
  - Evaluation metrics

Making your first data science program through Vibe Coding

prompt>_

"I would like to use the MT Samples dataset. Please download it from Hugging Face, load it into a pandas DataFrame, and display the first five rows."
（我想要使用MT Samples資料集, 請幫我從huggingface下載, 並且載入Dataframe裡, 輸出前五筆資料）

```python
import pandas as pd
from datasets import load_dataset

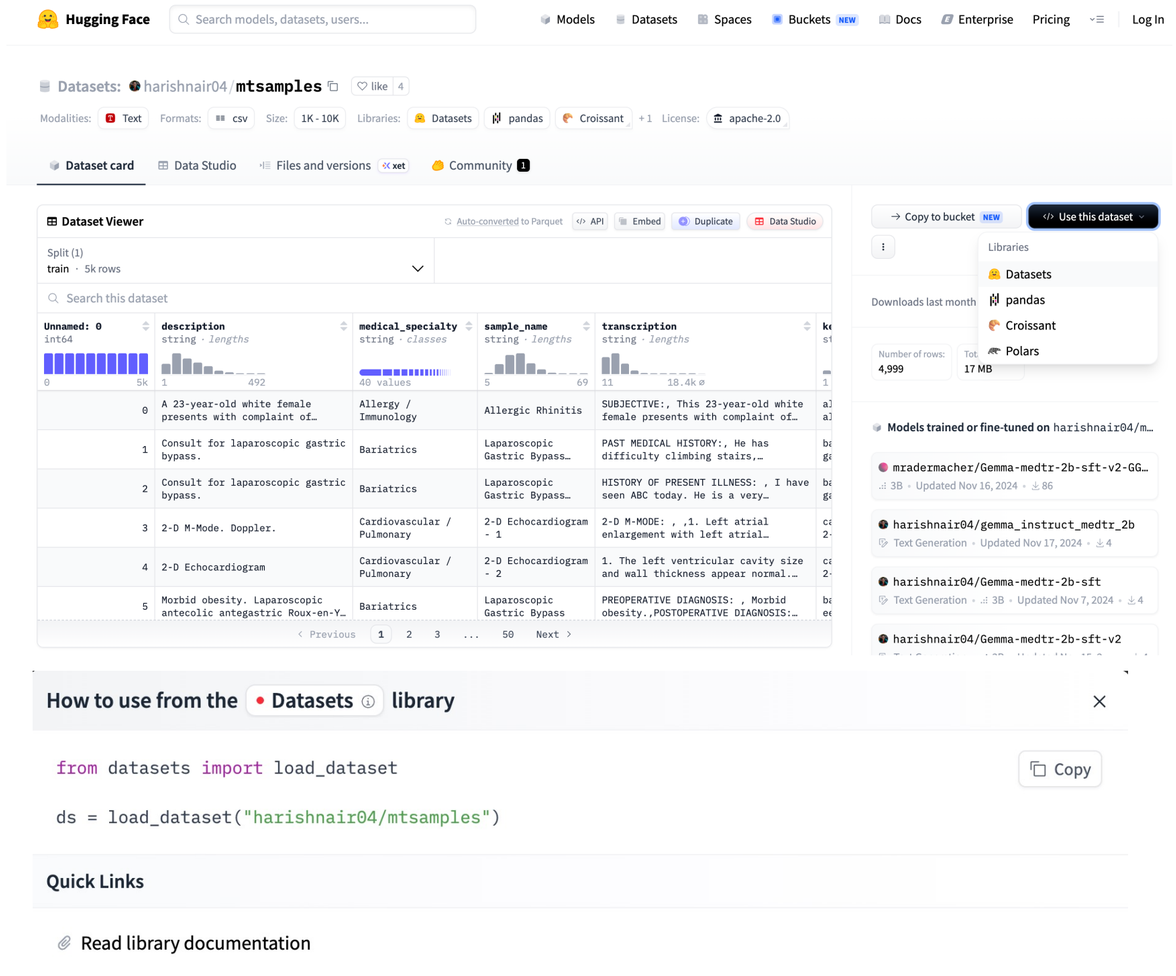
# Load the 'harishnair04/mtsamples' dataset from Hugging Face
dataset = load_dataset('harishnair04/mtsamples')

# Convert the 'train' split of the dataset to a pandas DataFrame
df_mt_samples = pd.DataFrame(dataset['train'])

# Display the first 5 rows of the DataFrame
display(df_mt_samples.head())
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fas
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN
README.md: 100% 31.0/31.0 [00:00<00:00, 609B/s]
mtsamples.csv: reconstructing file: 100% 17.0MB / 17.0MB, 1.68MB/s
mtsamples.csv: downloading bytes: 8.54MB, 844kB/s
Generating train split: 100% 4999/4999 [00:00<00:00, 16166.33 examples/s]
```

	Unnamed: 0	description	medical_specialty	sample_name	transcription	keywords
0	0	A 23–year–old white female presents with comp...	Allergy / Immunology	Allergic Rhinitis	SUBJECTIVE:, This 23–year–old white female pr...	allergy / immunology, allergic rhinitis, aller...
1	1	Consult for laparoscopic gastric bypass.	Bariatrics	Laparoscopic Gastric Bypass Consult – 2	PAST MEDICAL HISTORY:, He has difficulty climb...	bariatrics, laparoscopic gastric bypass, weigh...
2	2	Consult for laparoscopic gastric bypass.	Bariatrics	Laparoscopic Gastric Bypass Consult – 1	HISTORY OF PRESENT ILLNESS: , I have seen ABC ...	bariatrics, laparoscopic gastric bypass, heart...
3	3	2–D M–Mode. Doppler.	Cardiovascular / Pulmonary	2–D Echocardiogram – 1	2–D M–MODE: , ,1. Left atrial enlargement wit...	cardiovascular / pulmonary, 2–d m–mode, dopple...
4	4	2–D Echocardiogram	Cardiovascular / Pulmonary	2–D Echocardiogram – 2	1. The left ventricular cavity size and wall ...	cardiovascular / pulmonary, 2–d, doppler, echo...



Making your first data science program through Vibe Coding (cont.)

prompt>_

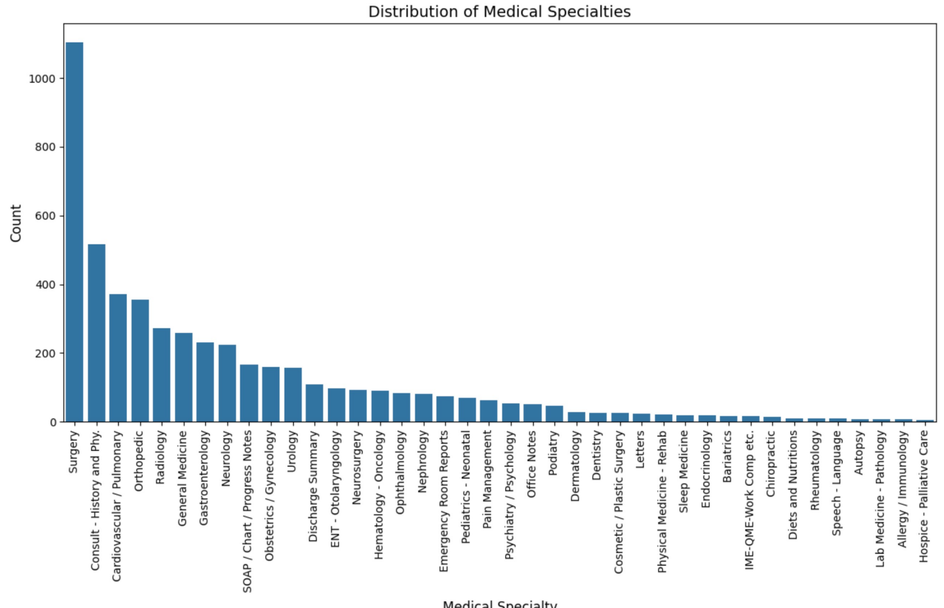
"Please calculate the number of unique values in the
（請幫我統計“medical_specialty”這個欄位有多少不同的值）

```
1 unique_specialties_count = df_mt_samples['medical_specialty'].nunique()
2 print(f"'medical_specialty' 欄位有 {unique_specialties_count} 個不同的值。")
```

```
'medical_specialty' 欄位有 40 個不同的值。
```

prompt>_

"Please visualize the distribution of values in the 'medical_specialty' column."
（請以視覺化的方式呈現“medical_specialty”這個欄位值的分布）




```
1 import matplotlib.pyplot as plt
2 import seaborn as sns
3
4 # Calculate the value counts of the 'medical_specialty' column
5 specialty_counts = df_mt_samples['medical_specialty'].value_counts()
6
7 # Set the figure size for better readability
8 plt.figure(figsize=(12, 8))
9
10 # Create a bar plot
11 sns.barplot(x=specialty_counts.index, y=specialty_counts.values)
12
13 # Add labels and title
14 plt.xlabel('Medical Specialty', fontsize=12)
15 plt.ylabel('Count', fontsize=12)
16 plt.title('Distribution of Medical Specialties', fontsize=14)
17
18 # Rotate x-axis labels to prevent overlap
19 plt.xticks(rotation=90)
20
21 # Adjust layout to make sure labels are not cut off
22 plt.tight_layout()
23
24 # Display the plot
25 plt.show()
```

WebCrawler

BeautifulSoup

- Beautiful Soup is a Python library designed for parsing HTML and XML documents. It provides tools for navigating, searching, and modifying the parse tree created from these documents, making it a popular choice for web scraping and data extraction tasks.
- **Parsing HTML and XML**:
  - Beautiful Soup can handle various types of web content, including "tag soup" (poorly structured HTML with unclosed tags or other errors), and build a parse tree from it.
- **Navigation and Searching**:
  - It offers intuitive methods to traverse the parse tree and locate specific elements based on their tags, attributes, CSS selectors, or text content.
- **Data Extraction**:
  - Once elements are identified, Beautiful Soup enables easy extraction of their text content, attribute values, or other relevant data.
- **Integration with Parsers**:
  - It works with various underlying HTML/XML parsers, such as html.parser (built-in Python), lxml, or html5lib.
- **Web Scraping**:
  - Beautiful Soup is frequently used in conjunction with libraries like requests (to fetch web pages) to automate the process of collecting data from websites.



Making your first data science program through Vibe Coding

prompt>_

Please use BeautifulSoup to scrape the table with class wikitable from:
https://en.wikipedia.org/wiki/List_of_countries_by_obesity_rate
Save the data into a DataFrame.

WIKIPEDIA
The Free Encyclopedia

Search Wikipedia | Search

Wiki Loves Monuments: Photograph a monument, help Wikipedia and win!
Learn more

List of countries by obesity rate

7 languages

Article | Talk | Read | Edit | View history | Tools

Contents hide

(Top)

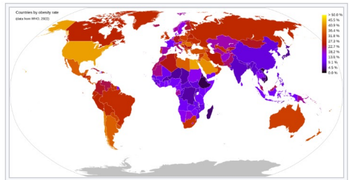
World Health Organization (2022 data)

See also

References

From Wikipedia, the free encyclopedia

This is a **list of countries by obesity rate**, with data from the World Health Organization (WHO), as of 2024. There are 191 registered countries on this ranking. This excludes Monaco, Liechtenstein, Vatican City and San Marino.



Countries by obesity rate, data from WHO 2022

World Health Organization (2022 data) [edit]

The data is from the World Health Organization.[1]

Rank	Country	Percentage of adults with obesity (BMI ≥ 30)
1	Tonga	70.5
2	Nauru	70.2
3	Tuvalu	63.9
4	Samoa	61.2
5	The Bahamas	47.6
6	Marshall Islands	47.3
7	Saint Kitts and Nevis	46.6
8	Kiribati	46.2
9	Micronesia	45.6
10	Kuwait	45.2

開始使用 AI 編寫或生成程式碼。

空白儲存格

請幫我用BeautifulSoup抓取
https://en.wikipedia.org/wiki/List_of_countries_by_obesity_rate
下wikitable這個表格中的資料，並且儲存成Dataframe

Gemini

```
[ ]  impo
     from
     impo

     url
     resp
     soup

     # Fi
     tabl

     # Extract data from the table
```

請幫我用BeautifulSoup抓取
https://en.wikipedia.org/wiki/List_of_countries_by_obesity_rate 下wikitable這個表格中的資料，並且儲存成Dataframe

空白儲存格

接受並執行 | 接受 | 取消

需要我幫忙建構什麼嗎?

Gemini 可能會出錯，因此請查證回覆內容，並謹慎使用程式碼。 瞭解詳情

Making your first data science program through Vibe Coding

```python
import requests
from bs4 import BeautifulSoup
import pandas as pd

url = "https://en.wikipedia.org/wiki/List_of_countries_by_obesity_rate"
response = requests.get(url)
soup = BeautifulSoup(response.text, "html.parser")

table = soup.find("table", {"class": "wikitable"})

data = []
for row in table.find_all("tr"):
    row_data = [cell.get_text(strip=True) for cell in row.find_all(["th", "td"])]
    data.append(row_data)

# Create a DataFrame from the extracted data
obesity_df = pd.DataFrame(data[1:], columns=data[0])

# Display the first few rows of the DataFrame
display(obesity_df.head())
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
/tmp/ipython-input-3975659098.py in <cell line: 0>()
     10
     11 data = []
---> 12 for row in table.find_all("tr"):
     13     row_data = [cell.get_text(strip=True) for cell in row.find_all(["th",
"td"])]
     14     data.append(row_data)

AttributeError: 'NoneType' object has no attribute 'find_all'
```

後續步驟： 說明錯誤

Gemini

```python
import requests
from bs4 import BeautifulSoup
import pandas as pd

url = "https://en.wikipedia.org/wiki/List_of_countries_by_obesity_rate"
-response = requests.get(url)
+headers = {'User-Agent': 'Mozilla/5.0 (DS-Classroom/1.0)'}
+response = requests.get(url, headers=headers)
soup = BeautifulSoup(response.text, "html.parser")

# Find all tables on the page
tables = soup.find_all("table", {"class": "wikitable"})
```

It seems like the code still can't find any tables with the class "wikitable". I'll print the HTML content of the page to help us figure out what the correct table identifier should be.

It looks like the website is blocking requests that don't have a proper User-Agent. I'll modify the code to include a User-Agent in the request headers, which should allow us to access the table data.

接受並執行 接受 取消

需要我幫忙建構什麼嗎?

Gemini 可能會出錯，因此請查證回覆內容，並謹慎使用程式碼。 瞭解詳情

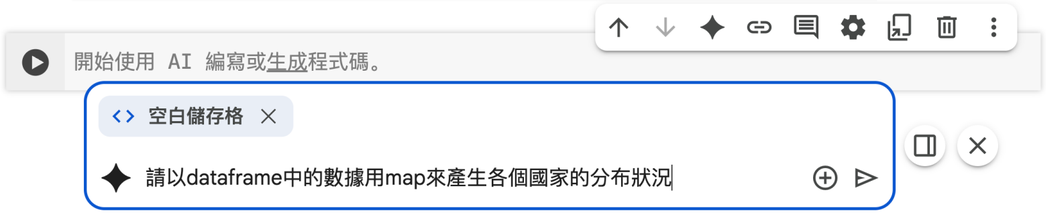
IndexError Traceback (most recent call last)

```python
[35] import requests

url = "https://en.wikipedia.org/wiki/List_of_countries_by_obesity_rate"
response = requests.get(url)
print(response.text)
```

```
Please set a user-agent and respect our robot policy https://w.wiki/4wJS. See also T400119.
```

Making your first data science program through Vibe Coding



prompt>_

Please use the data from the DataFrame to generate a map visualization showing the distribution across countries.

```
[41] import plotly.express as px

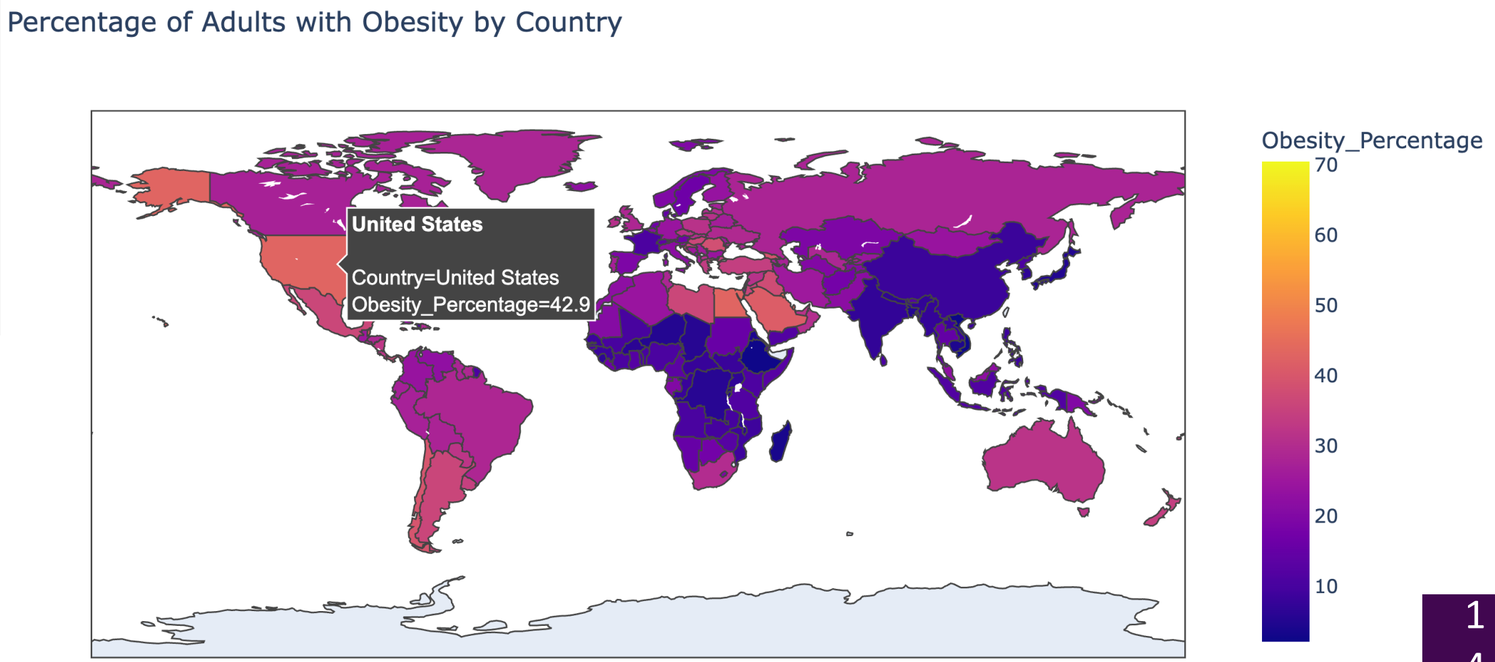
    # Rename the 'Percentage of adults with obesity (BMI≥ 30)' column for easier access
    obesity_df = obesity_df.rename(columns={'Percentage of adults with obesity (BMI≥ 30)': 'Obesity_Percentage'})

    # Convert the 'Obesity_Percentage' column to numeric, coercing errors
    obesity_df['Obesity_Percentage'] = pd.to_numeric(obesity_df['Obesity_Percentage'], errors='coerce')

    # Drop rows with NaN values in 'Obesity_Percentage' that resulted from coercion
    obesity_df.dropna(subset=['Obesity_Percentage'], inplace=True)

    # Create a world map choropleth visualization
    fig = px.choropleth(obesity_df,
                        locations="Country",
                        locationmode='country names',
                        color="Obesity_Percentage",
                        hover_name="Country",
                        color_continuous_scale=px.colors.sequential.Plasma,
                        title="Percentage of Adults with Obesity by Country")

    fig.show()
```



Discussion

Which regions cluster at the high end of obesity? Which at the low end?

How might income level, urbanization, or diet patterns relate to obesity rates?



Thanks for your listening



Q & A